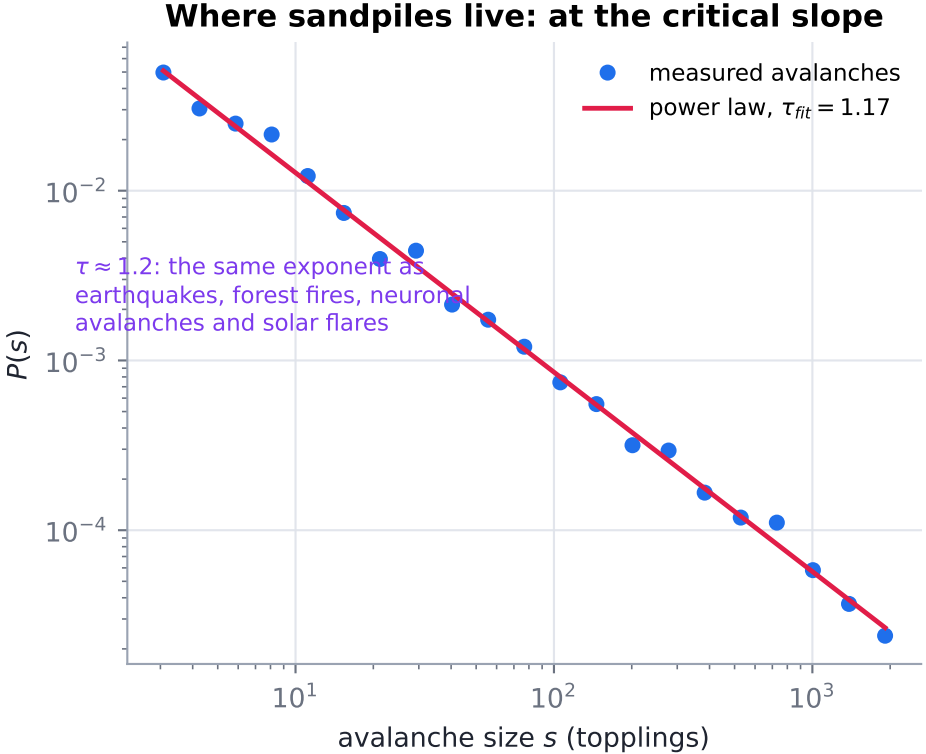
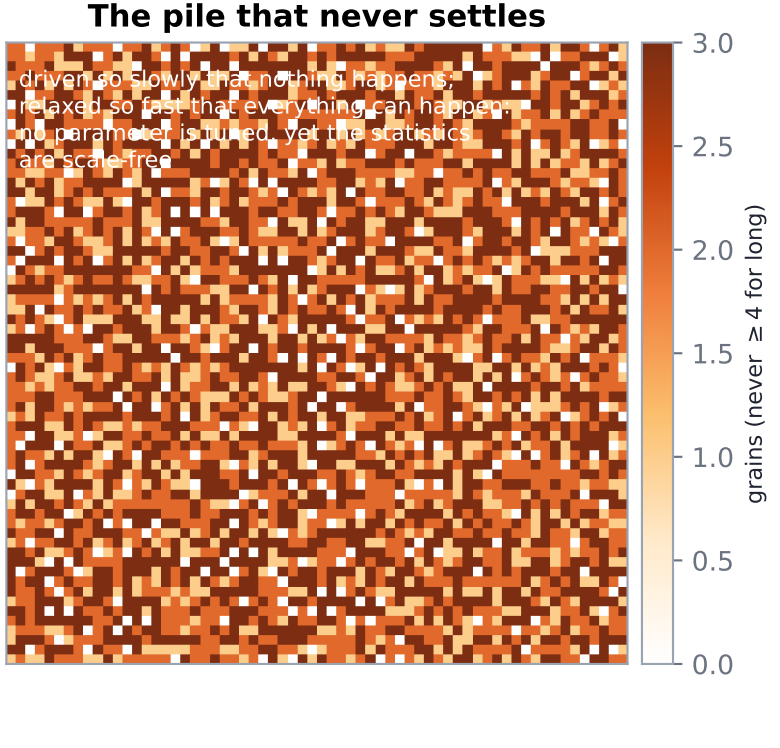

Where sandpiles live: at the critical slope
measured avalanches
power law, $\tau_{fit} = 1.17$
$\tau \approx 1.2$: the same exponent as earthquakes, forest fires, neuronal avalanches and solar flares
$P(s)$
10^{-2}
10^{-3}
10^{-4}
10^{1}
10^{2}
10^{3}
avalanche size s (toppling)


The pile that never settles
driven so slowly that nothing happens;
relaxed so fast that everything can happen;
no parameter is tuned, yet the statistics
are scale-free
grains (never ≥ 4 for long)
3.0
2.5
2.0
1.5
1.0
0.5
0.0